



HK IMO Training 2022-2023

Problem Set 2 Suggested Solutions



Problem 5 (Benelux MO 2014 Problem 1). Let a, b, c, d be positive integers. Find the smallest possible value of

$$S = \left\lfloor \frac{a+b+c}{d} \right\rfloor + \left\lfloor \frac{b+c+d}{a} \right\rfloor + \left\lfloor \frac{c+d+a}{b} \right\rfloor + \left\lfloor \frac{d+a+b}{c} \right\rfloor.$$

Here, $\lfloor x \rfloor$ denotes the largest integer not exceeding x .

Solution. Note that

$$\begin{aligned} S &> \left(\frac{a+b+c}{d} - 1 \right) + \left(\frac{b+c+d}{a} - 1 \right) + \left(\frac{c+d+a}{b} - 1 \right) + \left(\frac{d+a+b}{c} - 1 \right) \\ &= (a+b+c+d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) - 8 \\ &\geq 4^2 - 8 = 8 \end{aligned}$$

by the Cauchy-Schwarz inequality. Since S is an integer, we must have $S \geq 9$. On the other hand, when $a = 4$ and $b = c = d = 5$, we have $S = 9$. So the smallest possible value is 9. $\square$

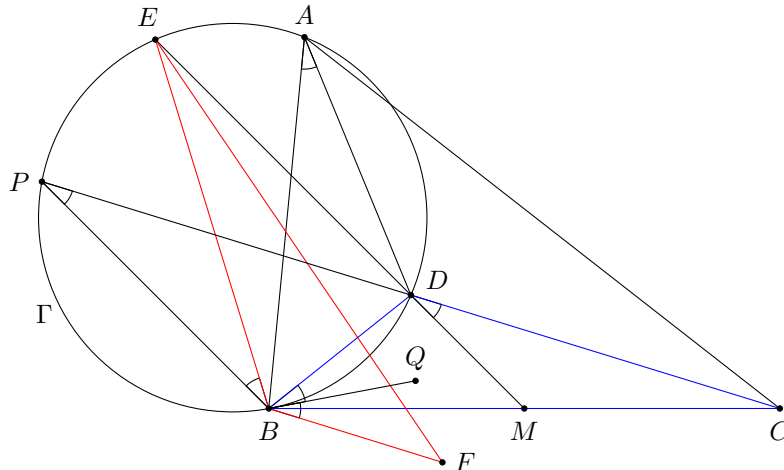
Problem 6 (ICO 2020 Problem 4). There are 99 people in a meeting. Some pairs of people shake hands with each other once. It is known that everyone has shaken hands for at least 81 times and at most 90 times. Prove that there are 10 people in the meeting who have shaken hands for the same number of times and there is another person who has shaken hands with all these 10 people.

Solution. We rephrase the question using graph theory, where vertices denote the people, and edges correspond to pairs of people who have shaken hands.

Let A be a vertex with the largest degree. If we partition its neighbours into groups $S_{81}, S_{82}, \dots, S_{90}$ such that each vertex V in S_i has degree i , then $|S_i| \leq 9$ for each i , or we are done. Since $\deg A \geq 81$, there are at least $\frac{81}{9} = 9$ nonempty groups. Thus, there must be a vertex with degree ≥ 89 . In particular, $\deg A \geq 89$. But then this shows there are at least $\left\lceil \frac{89}{9} \right\rceil = 10$ nonempty groups. Thus, there is a vertex with degree 90, and hence $\deg A = 90$. It follows that $|S_i| = 9$ for each i .

Consider the 9 vertices in S_{90} together with A . Each of them is not adjacent to 9 vertices. Since $99 > 10 \times 9$, there exists a vertex which is adjacent to all vertices in S_{90} and A . So we are done. $\square$

Problem 7 (Benelux MO 2012 Problem 3). In $\triangle ABC$, let M be the midpoint of BC . A variable point D lies inside $\triangle ABC$ such that $\angle BAD = \angle CDM$. Denote by Γ the circumcircle of $\triangle ABD$. Let $E \neq D$ be the intersection point of the line DM and Γ . Also, let F be the reflection of D in the tangent to Γ at B . Prove that the length of EF is independent of the position of D .



Solution. Let P be the second intersection point of the line CD and Γ , and let Q be a point on the tangent at B as shown.

Since $\angle CPB = \angle DPB = \angle DAB = \angle CDM$, we have $PB \parallel DM$. This shows $DEPB$ is an isosceles trapezoid. Also, as M is the midpoint of BC , D is the midpoint of PC . Now, we have $BD = BF$, $DC = DP = BE$ and

$$\angle BDC = \angle DBE + \angle EBP + \angle BPD = \angle DBE + 2\angle BPD = \angle DBE + 2\angle QBD = \angle DBE + \angle FBD = \angle FBE.$$

Therefore, $\triangle BDC \cong \triangle FBE$. This yields $BC = FE$, so the length of EF is independent of D . $\square$

Problem 8 (International Zhautykov Olympiad 2012 Problem 6). Solve the equation $2m^2 = n^{14} + 1$ in integers.

Solution. We have

$$2m^2 = n^{14} + 1 = (n^2 + 1)(n^{12} - n^{10} + n^8 - n^6 + n^4 - n^2 + 1).$$

Using long division, we find that $(n^2 + 1, n^{12} - n^{10} + n^8 - n^6 + n^4 - n^2 + 1) = (n^2 + 1, 7)$. As -1 is a quadratic nonresidue modulo 7, this greatest common divisor must be 1. Also, it is clear that $n^{12} - n^{10} + n^8 - n^6 + n^4 - n^2 + 1$ is odd. So we must have $n^2 + 1 = 2a^2$ and $n^{12} - n^{10} + n^8 - n^6 + n^4 - n^2 + 1 = b^2$ for some integers a and b .

WLOG assume $n \geq 0$. If $n \geq 3$, we have

$$256(n^{12} - n^{10} + n^8 - n^6 + n^4 - n^2 + 1) = (16n^6 - 8n^4 + 6n^2 - 5)^2 + (140n^4 - 196n^2) + 231 > (16n^6 - 8n^4 + 6n^2 - 5)^2$$

and

$$256(n^{12} - n^{10} + n^8 - n^6 + n^4 - n^2 + 1) = (16n^6 - 8n^4 + 6n^2 - 4)^2 - (32n^6 - 156n^4) - (208n^2 - 240) < (16n^6 - 8n^4 + 6n^2 - 4)^2.$$

This shows $(16n^6 - 8n^4 + 6n^2 - 5)^2 < (16b)^2 < (16n^6 - 8n^4 + 6n^2 - 4)^2$, which is impossible. Therefore, we must have $n = 0, 1, 2$. It is clear from $2m^2 = n^{14} + 1$ that n is odd. Therefore, $n = 1$ and $m^2 = 1$. The general solution is $(m, n) = (\pm 1, \pm 1)$. $\square$